

● HARD

● INFORMATION AND IDEAS

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04

10 questions · ~15 min

Q1

INFORMATION AND IDEAS

A member of the Otomi, an Indigenous people in Central Mexico, Octavio Medellín immigrated to the United States as a child, and his sculpture bears the impress of traditions on both sides of the border: US-based modernist sculpture, Mexican modernist painting, Otomi art, and the ancient sculpture of other Mexican Indigenous peoples, including the Maya. In his 1950 masterpiece *History of Mexico*, Medellín fuses these influences into a style so idiosyncratic that it resists efforts to view his work through the lens of nationality or cultural identity. Artists, he insisted, should strive for individual expression, even as they draw inspiration from their heritage and the communities where they live and work.

Which quotation from an art critic most directly challenges the underlined claim in the text?

- A. “Although a number of ancient Indigenous artistic traditions pictured human forms in profile, the forms populating the surface of *A History of Mexico* suggest a specifically Maya influence.”
- B. “In *A History of Mexico*, the synthesis of ancient and modernist traditions functions as a stylistic parallel to the work’s subject matter: a survey of centuries of Mexican history.”
- C. “Many critics focus on Indigenous influences in *A History of Mexico* and other key works by Medellín to the exclusion of influences from non-Indigenous art.”
- D. “While *A History of Mexico* features modernist motifs, it relies primarily on angular human forms in profile—a staple of Maya sculpture—and thus invites classification as Indigenous art.”

A team of biologists led by Jae-Hoon Jung, Antonio D. Barbosa, and Stephanie Hutin investigated the mechanism that allows *Arabidopsis thaliana* (thale cress) plants to accelerate flowering at high temperatures. They replaced the protein ELF3 in the plants with a similar protein found in another species (stiff brome) that, unlike *A. thaliana*, displays no acceleration in flowering with increased temperature. A comparison of unmodified *A. thaliana* plants with the altered plants showed no difference in flowering at 22° Celsius, but at 27° Celsius, the unmodified plants exhibited accelerated flowering while the altered ones did not, which suggests that _____ blank

Which choice most logically completes the text?

- A. temperature-sensitive accelerated flowering is unique to *A. thaliana*.
- B. *A. thaliana* increases ELF3 production as temperatures rise.
- C. ELF3 enables *A. thaliana* to respond to increased temperatures.
- D. temperatures of at least 22° Celsius are required for *A. thaliana* to flower.

The following text is from Milan Kundera's 1984 novel *The Unbearable Lightness of Being* (translated by Michael Henry Heim in 1984). Karenin is a dog that belongs to Tomas and Tereza.

Karenin was not overjoyed by the move to Switzerland [from Prague]. Karenin hated change. Dog time cannot be plotted along a straight line; it does not move on and on, from one thing to the next. It moves in a circle like the hands of a clock, which—they, too, unwilling to dash madly ahead—turn round and round the face, day in and day out following the same path. In Prague, when Tomas and Tereza bought a new chair or moved a flower pot, Karenin would look on in displeasure. It disturbed his sense of time. It was as though they were trying to dupe the hands of the clock by changing the numbers on its face.

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Which choice best states the main idea of the text?

- A. As a dog, Karenin possesses a sense of time that involves a strong preference for predictability and an aversion to disruption.
- B. After he's moved to a new home, Karenin's negative response to changes has become more pronounced.
- C. Similar to Tomas and Tereza, Karenin comprehends time as circular rather than as a straightforward progression.
- D. As is the case for other dogs, Karenin's sense of time seems to accelerate depending on the objects and places that surround him.

While attending school in New York City in the 1980s, Okwui Enwezor encountered few works by African artists in exhibitions, despite New York's reputation as one of the best places to view contemporary art from around the world. According to an arts journalist, later in his career as a renowned curator and art historian, Enwezor sought to remedy this deficiency, not by focusing solely on modern African artists, but by showing how their work fits into the larger context of global modern art and art history.

Which finding, if true, would most directly support the journalist's claim?

- A. As curator of the Haus der Kunst in Munich, Germany, Enwezor organized a retrospective of Ghanaian sculptor El Anatsui's work entitled *El Anatsui: Triumphant Scale*, one of the largest art exhibitions devoted to a Black artist in Europe's history.
- B. In the exhibition *Postwar: Art Between the Pacific and the Atlantic, 1945–1965*, Enwezor and cocurator Katy Siegel brought works by African artists such as Malangatana Ngwenya together with pieces by major figures from other countries, like US artist Andy Warhol and Mexico's David Siqueiros.
- C. Enwezor's work as curator of the 2001 exhibition *The Short Century: Independence and Liberation Movements in Africa, 1945–1994* showed how African movements for independence from European colonial powers following the Second World War profoundly influenced work by African artists of the period, such as Kamala Ibrahim Ishaq and Thomas Mukarobgwa.
- D. Enwezor organized the exhibition *In/sight: African Photographers, 1940 to the Present* not to emphasize a particular aesthetic trend but to demonstrate the broad range of ways in which African artists have approached the medium of photography.

Among social animals that care for their young, such as chickens, macaque monkeys, and humans, newborns appear to show an innate attraction to faces and face-like stimuli. Elisabetta Versace and her colleagues used an image of three black dots arranged in the shape of eyes and a nose or mouth to test whether this trait also occurs in *Testudo* tortoises, which live alone and do not engage in parental care. They found that tortoise hatchlings showed a significant preference for the image, suggesting that _____ blank

Which choice most logically completes the text?

- A. face-like stimuli are likely perceived as harmless by newborns of social species that practice parental care but as threatening by newborns of solitary species without parental care.
- B. researchers should not assume that an innate attraction to face-like stimuli is necessarily an adaptation related to social interaction or parental care.
- C. researchers can assume that the attraction to face-like stimuli that is seen in social species that practice parental care is learned rather than innate.
- D. newly hatched *Testudo* tortoises show a stronger preference for face-like stimuli than adult *Testudo* tortoises do.

Poetry in Classical Nahuatl, the language of the Aztec Empire, relies on *difrasismo*, or a parallel noun construction that conventionally operates as a single metaphor. For example, the common difrasismo *in cuauhtli in ocelotl* (literally, “the eagle, the jaguar”) signifies “warrior.” The device’s function is both formal—providing structure to lines of verse—and ritual: semantic relations among the two nouns and the concept they signify can be tenuous, as in the previous example, such that difrasismos are often only intelligible according to the conceptual associations observed in Aztec ceremonial culture.

Which statement about the difrasismo *in cuauhtli in ocelotl* is most strongly supported by the text?

- A. Its metaphorical significance derives from the semantic equivalence of the two nouns constituting the difrasismo.
- B. Its unintelligibility may cause its formal function within a line of verse to go unnoticed by present-day readers.
- C. Its apparent obscurity can be resolved when considered in the proper cultural context.
- D. Its frequency in Classical Nahuatl poetry confirms its intelligibility to the Aztec audience.

The linguistic niche hypothesis (LNH) posits that the exotericity of languages (how prevalent non-native speakers are) and grammatical complexity are inversely related, which the LNH ascribes to attrition of complex grammatical rules as more non-native speakers adopt the language but fail to acquire those rules. Focusing on two characteristics that are positive indices of grammatical complexity, fusion (when new phonemes arise from the merger of previously distinct ones) and informativity (languages' capacity for meaningful variation), Olena Shcherbakova and colleagues conducted a quantitative analysis for more than 1,300 languages and claim the outcome is inconsistent with the LNH.

Which finding, if true, would most directly support Shcherbakova and colleagues' claim?

- A. Shcherbakova and colleagues' analysis showed a slightly negative correlation between grammatical complexity and fusion and between grammatical complexity and informativity.
- B. Shcherbakova and colleagues' analysis showed a slightly negative correlation between grammatical complexity and exotericity.
- C. Shcherbakova and colleagues' analysis showed a slightly positive correlation between grammatical complexity and fusion.
- D. Shcherbakova and colleagues' analysis showed a slightly positive correlation between fusion and exotericity and between informativity and exotericity.

Data collected by the Mars rover *Curiosity* at the Gale Crater's Murray Formation are suggestive of hydrological deposition of sediment in the distant past. To characterize the nature of the depositional environment, Frances Rivera-Hernández et al. analyzed the grain size of Murray Formation sediment, finding that although there are intervals of coarse grains, most of the sediment consists of fine grains that show signs of cracking due to episodic desiccation. Rivera-Hernández et al. concluded that the coarse grains are sandstone, which tends to be deposited by flowing water, whereas the fine grains are mudstone, which is slowly deposited by settling out of suspension in low-flow water, leading the researchers to posit that _____ blank

Which choice most logically completes the text?

- A. although the area of the Murray Formation experienced a prolonged period of dryness that prevented a lake from forming, water flowing from a distant source was present.
- B. a lake existed at the Murray Formation for a prolonged period, though the lake occasionally experienced drying and there were periods in which one or more streams were present.
- C. one or more streams existed at the Murray Formation for an extended period until being replaced by a lake that persisted for only a brief period before permanently drying.
- D. a stream-fed lake was present at the Murray Formation for an extended period, and although the streams experienced occasional drying, the lake did not.

The following text is from William Shakespeare's play *The Tempest*, first performed in 1611. Miranda has lived on an island with her father, Prospero, since she was three years old. Prospero has stated that Miranda likely does not remember anything other than her life on the island.

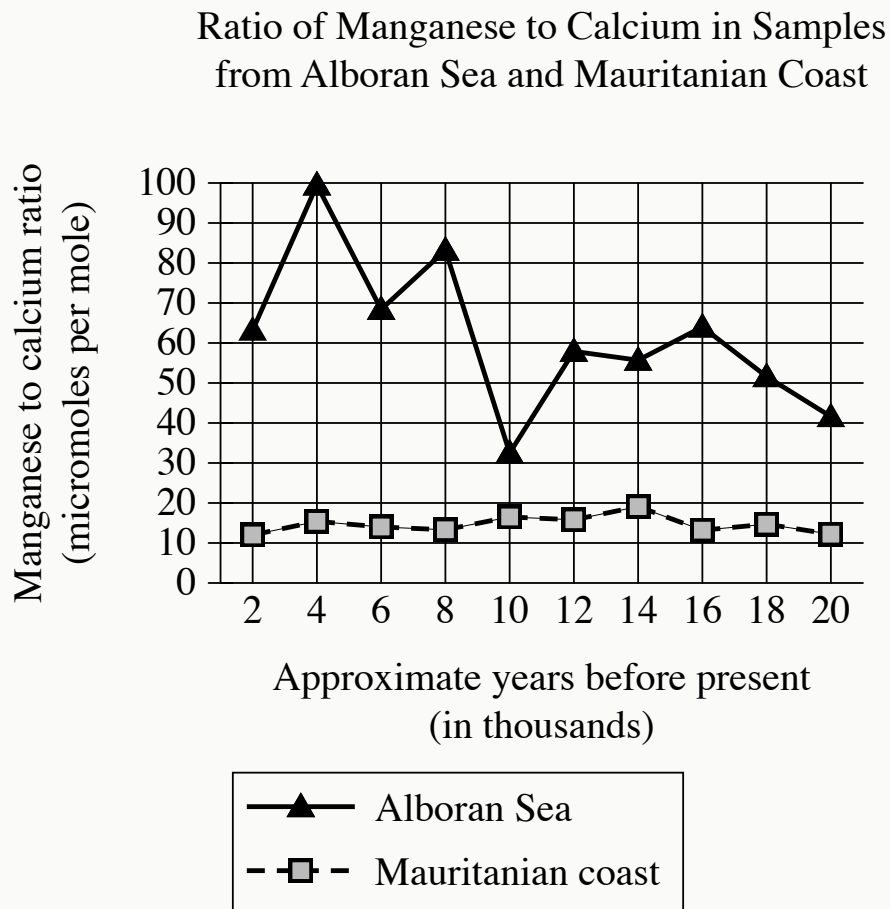
MIRANDA: 'Tis far off,
And rather like a dream than an assurance
That my remembrance warrants. Had I not
Four or five women once that tended me?

PROSPERO: Thou hadst, and more, Miranda. But how is it
That this lives in thy mind? What seest thou else
In the dark backward and abysm of time?
If thou remember'st ought ere thou camest here,
How thou camest here thou mayst.

In the text, which point does Prospero most directly make about Miranda and her memories?

- A. Miranda's reminiscences about her early childhood have a melancholy quality that betrays her discontented view of her current circumstances.
- B. Miranda's doubts about the accuracy of one recollection of a place other than the island are clouding her judgment and seem to be making her reluctant to explore her recollection of traveling to the island.
- C. Miranda's ability to summon details of an experience she had before arriving on the island suggests that she may also be able to summon details of her arrival on the island.

- D. Miranda's impression of a scene is vague because she is remembering a scenario she had daydreamed about as a child rather than a scenario that had occurred in reality.



- The following 2 lines are shown:
 - Alboran Sea
 - Mauritanian coast
- The Alboran Sea line:
 - Begins at 2 thousand, 63.2 micromoles per mole
 - Rises sharply to 4 thousand, 99.5 micromoles per mole
 - Falls sharply to 6 thousand, 68.4 micromoles per mole
 - Rises sharply to 8 thousand, 83.2 micromoles per mole
 - Falls sharply to 10 thousand, 32.5 micromoles per mole
 - Rises sharply to 12 thousand, 57.9 micromoles per mole
 - Falls gradually to 14 thousand, 55.7 micromoles per mole
 - Rises gradually to 16 thousand, 64 micromoles per mole

- Falls sharply to 18 thousand, 51.6 micromoles per mole
- Falls sharply to 20 thousand, 41.6 micromoles per mole
- The Mauritanian coast line:
 - Begins at 2 thousand, 12 micromoles per mole
 - Rises gradually to 4 thousand, 15.4 micromoles per mole
 - Falls gradually to 6 thousand, 14 micromoles per mole
 - Falls gradually to 8 thousand, 13.3 micromoles per mole
 - Rises gradually to 10 thousand, 16.5 micromoles per mole
 - Falls gradually to 12 thousand, 15.8 micromoles per mole
 - Rises gradually to 14 thousand, 19.1 micromoles per mole
 - Falls gradually to 16 thousand, 13.2 micromoles per mole
 - Rises gradually to 18 thousand, 14.7 micromoles per mole
 - Falls gradually to 20 thousand, 12.2 micromoles per mole

The population of the coral *Lophelia pertusa* declined significantly around 9,000 years ago in the Alboran Sea and around 11,000 years ago near the Mauritanian coast. Using the ratio of manganese to calcium, which inversely correlates with ocean oxygenation levels, marine scientist Rodrigo da Costa Portilho-Ramos and colleagues evaluated whether oxygenation played a role in the declines of *L. pertusa*. The researchers concluded that oxygenation may have been important in the Alboran Sea but not near the Mauritanian coast, since _____ blank

Which choice most effectively uses data from the graph to complete the statement?

- A. a substantial increase in oxygenation in the Alboran Sea corresponded with the local decline in *L. pertusa*, but the opposite relationship between oxygenation and *L. pertusa* was found near the Mauritanian coast.
- B. *L. pertusa* declined in the Alboran Sea during a period of substantial local decline in oxygenation, but *L. pertusa* declined near the Mauritanian coast during a period of little local change in oxygenation.
- C. oxygenation in the Alboran Sea was higher before the decline in *L. pertusa* than after the decline, whereas oxygenation near the Mauritanian coast was relatively low both before and after the decline in *L. pertusa*.

- D. oxygenation in the Alboran Sea tended to be substantially higher than oxygenation near the Mauritanian coast during the period studied.